

Aloïs Dauger

PhD Student in Mathematical Biology, Sorbonne University

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Research Interests

I am currently a PhD student under the supervision of Hedi Soula and Chloé Audebert. My work focuses on the development of mathematical models to describe the size distribution of adipocytes and their population dynamics, with a quantitative and integrative approach to biological phenomena. I am particularly interested in the biological interpretation of mathematical models applied to tissue functions and physiological systems.

Keywords: Mathematical modeling of biological systems, Computational biology, Ordinary Differential Equations, Metabolic diseases, Adipose tissue dynamics.

Education

Master's Degree (M2) – Integrative Biology and Physiology Modeling and Data Analysis (MAD) Sorbonne University, Paris	2022–2023
Master's Degree (M1) – Integrative Biology and Physiology Systems Biology Sorbonne University, Paris	2021–2022
Bachelor's Degree in Life Sciences Sorbonne University, Paris	2019–2021
Preparatory Classes (BCPST) Lycée Albert Schweitzer, Le Raincy	2017–2019

Research Experience

PhD Student – Jacques-Louis Lions Laboratory, Paris <i>Mathematical modeling of adipocyte size distribution and population dynamics</i> • Development of mathematical models for adipose tissue dynamics • Analysis of adipocyte size distributions using ODE-based frameworks • Integration of biological interpretation with quantitative modeling	2023–Present
Master's Thesis – Nutriomics Laboratory, Paris <i>Modeling adipocyte size dynamics during weight gain</i> • Developed stochastic differential equation models of lipid exchange • Studied hypertrophy versus hyperplasia in adipose tissue growth • Analyzed lipid flux variability	2023

Master’s Internship – MAP5 Laboratory, Paris 2022
Inference of CD8 T-cell differentiation trajectories

- Applied optimal transport to biological datasets
- Implemented and extended Waddington-OT
- Generated synthetic datasets for cell differentiation

Bachelor’s Internship – Nutriomics Laboratory, Paris 2021
Modeling adipocyte size distributions

- Solved systems of ordinary differential equations numerically
- Investigated bimodal distributions

Research Observation Internship – LOCEAN Laboratory, Paris 2014

Publications

- Louis Fostier, **Aloïs Dauger**, Romain Yvinec, Magali Ribot, Chloe Audebert, Hedi Soula (2026). *Rapid cell turnover to model adipocyte size distribution*. *Journal of Theoretical Biology*.
- **Aloïs Dauger**, Hedi Soula, Chloe Audebert (2025). *Adipocyte size distribution: Mathematical model of a tissue property*. *Mathematical Biosciences*.

Teaching Experience

- Mathematics for Biology – Undergraduate Year 2 (L2)
- Statistics for Biology – Undergraduate Year 2 (L2)
- Mathematics for Biology – Undergraduate Year 3 (L3)
- Statistics for Biology – Undergraduate Year 3 (L3)
- Python for Physiology – Master’s Year 1 (M1)

Supervision

Elodie Husson – Master’s Year 1 intern Apr–Jun 2025
Nour Boughalia – High school intern (Terminale) Apr–May 2024

Collective Responsibilities

Psycho-social Risks Advisor
 Nutriomics Laboratory 2023–2026

PhD Student Representative, Lab Council Member
 Jacques-Louis Lions Laboratory 2023–2024

Communications

- **CMDO/AFERO Seminar** – Paris, France, Jan 2026
 Oral presentation

- **Global Health Winter School (UNAM–Sorbonne University)** – Mexico, Dec 2025
Oral presentation
- **Journées Maths Bio Santé (JMBS 2025)** – Montpellier, France, Nov 2025
Oral presentation
- **Doctoral School Days (JED-ED394)** – Paris, France, Jun 2025
Poster presentation — *Best Poster Award*
- **MATIDAYS: Mathematical Models of Lipid Storage** – Paris, France, Nov 2024
Oral presentation
- **European Conference on Mathematical and Theoretical Biology (ECMTB 2024)** – Toledo, Spain
Poster presentation — *Best Poster Award*
- **Journées Maths Bio Santé (JMBS 2024)** – Nantes, France, Jun 2024
Poster presentation
- **Doctoral School Days (JED-ED394)** – Paris, France, Jun 2024
Poster presentation
- **Association Française d’Étude et de Recherche sur l’Obésité (AFERO 2024)** – Nantes, France, Jan 2024
Poster presentation — *Best Poster Award*
- **Journées Maths Bio Santé (JMBS 2023)** – Marne-la-Vallée, France, Nov 2023
Poster presentation

Technical Skills

Programming: Python, MATLAB, R

Mathematical Modeling: ODEs, SDEs, agent-based modeling

Data Analysis: Clustering, machine learning, single-cell RNA-seq analysis

Languages

French (native), English (C1), Spanish (B1)

Honors & Awards

Best Poster Award – JED-ED394 2025

Best Poster Award – ECMTB 2024

Best Poster Award – AFERO 2024